

ESLAM TAREK NABAWY

Flutter Software Engineer | Mobile & Desktop Apps | Firebase | WebRTC | AI Automation

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PROFESSIONAL SUMMARY

Flutter Software Engineer and Computer Science graduate with 3+ years of hands-on freelance and project experience building cross-platform mobile and desktop applications. Strong in Dart, Flutter, Firebase, REST APIs, WebRTC, Riverpod, BLoC/Cubit, Drift, Hive, Git, and n8n/LLM automation. Built real products across Android, Windows, Firebase-backed workflows, local-first desktop systems, and private peer-to-peer communication.

TECHNICAL SKILLS

Programming Languages: Dart, JavaScript, Python, C++, Java, HTML, CSS

Mobile & Frontend: Flutter, Flutter Desktop, Android, Windows desktop, responsive UI, Figma, React.js familiarity

State Management & Architecture: Riverpod, BLoC, Cubit, MVC, GoRouter, repository pattern, feature-based architecture

Backend, APIs & Data: Firebase Auth, Firebase Realtime Database, Cloud Firestore, Firebase Storage, REST API integration, Drift, Hive, SQL, NoSQL

Realtime & Platform: WebRTC data channels, voice calls, video calls, file transfer flows, local persistence, release artifacts

AI & Automation: n8n workflow automation, OpenAI API, Claude API, LLM API integration, webhooks, AI-assisted app creation

Tools & Engineering: Git, GitHub, GitHub Actions, CI/CD concepts, unit testing, debugging, OOP, design patterns, data structures, algorithms, Docker familiarity, AWS familiarity

PROJECTS

Brox | Offline-First Workspace Operations App | Flutter Desktop, Hive, Cubit/Bloc

- Built a local-first Flutter desktop operations system for coworking/front-desk workflows including check-in, checkout, live sessions, customers, rooms, desks, reservations, kitchen inventory, finance, staff, memberships, analytics, roles, audit logs, export, restore, and app health.
- Designed the app around GoRouter shell routing, Cubit/Bloc state management, Hive CE local persistence, JSON-backed repositories, EasyLocalization, shared UI primitives, Windows release scripts, and GitHub Actions.
- Maintained production-style project structure with 531 Dart files under lib/src, 98 Dart test files, 520 widgets, local schema version 3, backup/restore flows, permissions, diagnostics, and release documentation.

Rain | Private P2P Messenger | Flutter, WebRTC, Firebase, Drift

- Built a private peer-to-peer chat app for Android and Windows with accepted friendships, WebRTC data-channel messaging, connection diagnostics, one-to-one file transfer, voice calls, and video calls.
- Separated Flutter UI, Riverpod state, Drift local persistence, Firebase Auth/Realtime Database signaling, Remote Config, session policy, and WebRTC media/data primitives across app, core, protocol, and peer packages.
- Implemented username sign-in, friend search, friend requests, blocking, call/file conflict guards, direct/relay route visibility, recovery states, and Android/Windows release artifacts.

CampusSuit | Campus Life Management App | Graduation Project | Flutter, Firebase

- Built a Flutter campus app covering onboarding, campus activities, events, profile management, live data, Firebase-backed workflows, and REST API integrations.
- Structured feature screens, state management, and data-backed UI flows so graduation-project requirements could grow without turning the UI layer into backend glue.

AI Workflow Automation | n8n, OpenAI API, Claude API, Webhooks

- Built n8n workflows that connect OpenAI/Claude APIs, webhooks, notifications, and external services into repeatable AI automation pipelines.
- Automated data processing, AI-generated content flows, smart notifications, conditional routing, and error-aware workflow paths.

So She Picks | E-Commerce Ordering App | Flutter, Firebase

- Delivered responsive ordering UI, Firebase-backed order management, checkout flow, and loyalty points behavior for a mobile commerce experience.

PROFESSIONAL EXPERIENCE

Freelance Software Developer | Self-Employed | 6th October City, Egypt | Nov 2022 - Present

- Built Flutter mobile and desktop applications using Dart, Firebase, REST APIs, responsive UI, local persistence, and feature-based architecture across client and personal products.
- Integrated Firebase Auth, Firestore, Realtime Database, Firebase Storage, REST APIs, routing, and state management with BLoC/Cubit and Riverpod-style patterns.
- Delivered product workflows from requirements to release artifacts using Git/GitHub, documentation, debugging, and AI-assisted development while keeping implementation decisions explicit and reviewable.

- Created n8n and LLM API workflows using OpenAI, Claude, webhooks, conditional routing, and notifications to automate repetitive product and content operations.

Programming Instructor | Local School & Private Sessions | 6th October City, Egypt | Aug 2023 - Present

- Taught programming fundamentals, OOP, algorithms, data structures, debugging, and practical software development through structured exercises.
- Mentored students through problem decomposition, implementation habits, and code-review thinking.

LEADERSHIP AND VOLUNTEERING

Flutter Head | Google Developer Student Club | Oct 2022 - Oct 2023

- Led Flutter learning sessions, mentored beginner developers, and helped organize community technical events.
- Coordinated NextGen and Google I/O extended events with the GDSC AEA team.

ICPC Core Team Member | Competitive Programming | Jun 2020 - Oct 2023

- Organized problem-solving contests and supported new members in algorithmic thinking and contest preparation.
- Built a three-year habit of solving difficult problems under time pressure and explaining solutions clearly.

EDUCATION

Bachelor of Science in Computer Science | AEA Academy, 6th October City, Egypt | Oct 2020 - Jul 2024

LANGUAGES

Arabic: Native | English: Upper-Intermediate / Professional proficiency